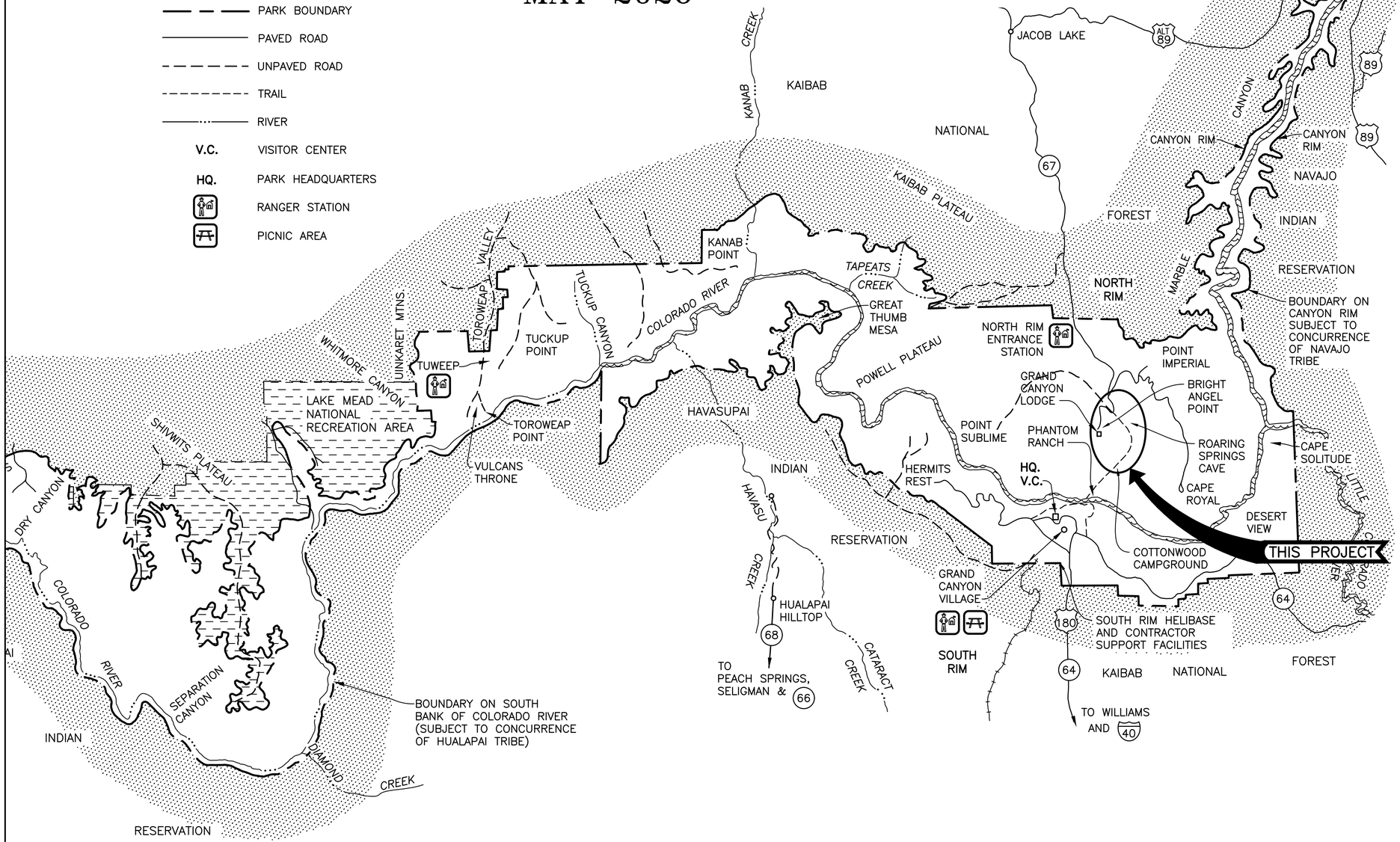
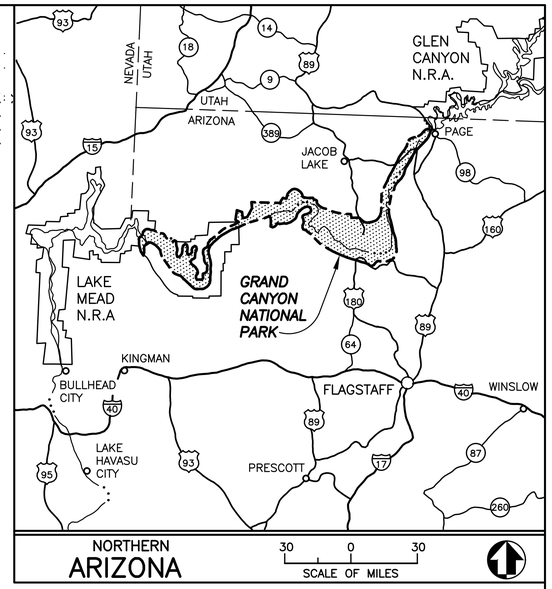
GRAND CANYON NATIONAL PARK

WORLD HERITAGE SITE

REHABILITATE NORTH RIM INFRASTRUCTURE

MAY 2026





SCALE OF MILES
5 0 5 10
5 0 5 10 15
SCALE OF KILOMETERS

PMIS – 318719

BASIC DATA: LAND STATUS MAP 113/80,011 7/75; PARK MINI FOLDER 1989; REVISED & REDRAWN 2/91. UPDATED 11/98. SGS

A/E FIRM	Mark	Sheet	REVISION	Date	Initial
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202					
	▽		CONSTRUCTION DOCUMENTS	MAY 2026	

REGISTERED PROFESSIONAL ENGINEER — CERTIFICATE NO. 41909 — RYAN RHOADES — Date Signed 5-21-26 — ARIZONA, U.S.A.

QUALITY DESIGN CERTIFICATION

☐ Prepared in Accordance with Design Development (Title I) ____ Drawing No.
OR
☐ Variance from Design Development (Title I) Approved by Superintendent on ____ Date
OR
☐ Construction Drawing Not Preceded by Design Development (Title I)

Project Manager ____ Date

CONSTRUCTION DOCUMENTS

UNITED STATES
DEPARTMENT OF THE INTERIOR

NATIONAL PARK SERVICE
DENVER SERVICE CENTER

TITLE OF DRAWING
COVER SHEET
LOCATION WITHIN PARK
NORTH RIM AND INNERCANYON
NAME OF PARK
GRAND CANYON NATIONAL PARK

REGION	COUNTY	STATE
INTERMOUNTAIN	COCONINO	ARIZONA

DRAWING NO. 113 182752
PKG. NO. 318719
SHEET 1 OF 98

4/29/2026 Lynch, Nicole C:\Users\nlynch\DC\ACCDocs\HDR\103505545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\GEN\000-G001.dwg

GRAND CANYON NATIONAL PARK
WORLD HERITAGE SITE
REHABILITATE NORTH RIM INFRASTRUCTURE
VOLUME 0 – GENERAL

VOLUME #	DESCRIPTION
VOLUME 0	GENERAL
VOLUME 1	PIPELINES - NORTH RIM
VOLUME 2	PIPELINES - NORTH RIM TO ROARING SPRINGS PUMPHOUSE (RSPH)
VOLUME 3	PIPELINES - NORTH RIM TO SUPAI TUNNEL
VOLUME 4	PIPELINES - ROARING SPRINGS CAVE TO ROARING SPRINGS PUMPHOUSE (RSPH)
VOLUME 5	ROARING SPRINGS PUMPHOUSE (RSPH)
VOLUME 6	NORTH RIM WATER TREATMENT PLANT
VOLUME 7	NOT USED
VOLUME 8	NORTH RIM WASTEWATER COLLECTION SYSTEM
VOLUME 9	POWER DISTRIBUTION - NORTH RIM
VOLUME 10	POWER DISTRIBUTION - NORTH RIM TO ROARING SPRINGS PUMPHOUSE (RSPH)
VOLUME 11	POWER DISTRIBUTION - ROARING SPRINGS PUMPHOUSE (RSPH) TO MANZANITA
VOLUME 12	POWER SYSTEM - COTTONWOOD CAMPGROUND

SHEET	SUB SHEET	SUB AREA	TITLE OF SHEET
1	000G001	GENERAL	COVER SHEET
2	000G002	GENERAL	SHEET INDEX
3	000G003	GENERAL	ABBREVIATIONS
4	000G004	GENERAL	GENERAL LEGEND
5	000G005	GENERAL	PROJECT AREAS AND VICINITY MAP
6	000G102	GENERAL	NORTH RIM LIMITS OF CONSTRUCTION
7	000G103	GENERAL	INNER CANYON LIMITS OF CONSTRUCTION
8	000G105	GENERAL	NORTH RIM CONSTRUCTION CAMP VICINITY MAP
9	000G106	GENERAL	NORTH RIM WWTP CONSTRUCTION TRAILERS AND HELICOPTER STAGING AREA
10	000G107	GENERAL	NORTH RIM CSF HELICOPTER STAGING AREA (SOUTH RIM)
11	000G108	GENERAL	NORTH RIM LINDBERG HILL CONSTRUCTION CAMP
12	000G109	GENERAL	NOT USED
13	000G110	GENERAL	NORTH RIM COTTONWOOD CAMPGROUND CONSTRUCTION CAMP
14	000G111	GENERAL	NORTH RIM NORTON CT CAMPGROUND CONSTRUCTION CAMP
15	001G120	GENERAL	CONTRACTOR STAGING AND ACCESS PLAN WTP
16	001G122	GENERAL	CONTRACTOR STAGING AND ACCESS PLAN BALLFIELD SITE
17	002C001	GENERAL	GENERAL NOTES, ABBREVIATIONS, AND LEGEND CIVIL
18	002C101	GENERAL	SURVEY POTHOLE PLAN (NORTH RIM)
19	002C102	GENERAL	SURVEY POTHOLE TABLE (NORTH RIM)
20	002C103	GENERAL	SURVEY CONTROL (NORTH RIM)
21	002C104	GENERAL	SURVEY CONTOL POINT TABLE (NORTH RIM)
22	002C105	GENERAL	OVERALL SURVEY CONTROL PLAN (INNER CANYON)
23	002C501	GENERAL	CIVIL DETAILS UTILITIES
24	002C502	GENERAL	CIVIL DETAILS - UTILITIES CONTINUED
25	002C503	GENERAL	CIVIL DETAILS - UTILITIES CONTINUED
26	002C504	GENERAL	CIVIL DETAILS - UTILITIES CONTINUED
27	002C505	GENERAL	CIVIL DETAILS - JOINTS
28	002C506	GENERAL	CIVIL DETAILS - PAVING
29	002C507	GENERAL	CIVIL DETAILS - GRADING
30	002C508	GENERAL	EROSION AND SEDIMENT CONTROL NOTES
31	002C509	GENERAL	CIVIL DETAILS - GESC CONTINUED
32	002C510	GENERAL	CIVIL DETAILS - GESC CONTINUED
33	002C511	GENERAL	CIVIL DETAILS - GESC CONTINUED
34	002C512	GENERAL	CIVIL DETAILS - GESC CONTINUED
35	003S501	GENERAL	STRUCTURAL DETAILS - PADS
36	003S502	GENERAL	STRUCTURAL DETAILS - ANCHORING
37	003S503	GENERAL	STRUCTURAL DETAILS - JOINTS
38	003S504	GENERAL	STRUCTURAL DETAILS - REINFORCING
39	003S505	GENERAL	STRUCTURAL DETAILS - CONNECTIONS
40	003S506	GENERAL	STRUCTURAL DETAILS - GENERAL
41	003S507	GENERAL	STRUCTURAL DETAILS - GENERAL CONTINUED
42	004A501	GENERAL	ARCHITECTURAL DETAILS - GENERAL
43	004A502	GENERAL	ARCHITECTURAL DETAILS - GENERAL
44	004A503	GENERAL	ARCHITECTURAL DETAILS - GENERAL
45	004A504	GENERAL	ARCHITECTURAL DETAILS - DOOR FRAMES
46	004A505	GENERAL	ARCHITECTURAL DETAILS - DOOR FRAMES CONTINUED
47	004A506	GENERAL	ARCHITECTURAL DETAILS - ALUMINUM WINDOW FRAMES
48	004A507	GENERAL	ARCHITECTURAL DETAILS - HATCHES AND MISCELLANOUS
49	004A508	GENERAL	ARCHITECTURAL DETAILS - RAILINGS
50	004A509	GENERAL	ARCHITECTURAL DETAILS - WALL TYPES
51	004A510	GENERAL	ARCHITECTURAL DETAILS - MOUNTING HEIGHTS
52	004A511	GENERAL	ARCHITECTURAL DETAILS - INTERIORS
53	004A512	GENERAL	ARCHITECTURAL DETAILS - INTERIORS CONTINUED
54	004A601	GENERAL	ARCHITECTURAL DOOR SCHEDULES
55	004A602	GENERAL	ARCHITECTURAL FINISH SCHEDULES
56	004A603	GENERAL	ARCHITECTURAL FINISH LEGEND & NOTES
57	005D001	GENERAL	GENERAL NOTES, ABBREVIATIONS, AND LEGEND PROCESS
58	005D101	GENERAL	NORTH RIM RW HYDRAULIC PROFILE AND VALVE SCHEDULE
59	005D102	GENERAL	NORTH RIM FW HYDRAULIC PROFILE AND VALVE SCHEDULE
60	005D103	GENERAL	SUPAI WATER HYDRAULIC PROFILE AND VALVE SCHEDULE
61	005D501	GENERAL	PROCESS DETAILS - PIPE SUPPORTS
62	005D502	GENERAL	PROCESS DETAILS - PIPE SUPPORTS CONTINUED
63	005D503	GENERAL	PROCESS DETAILS - PIPES AND APPURTENANCES
64	005D504	GENERAL	PROCESS DETAILS - MISCELLANEOUS
65	005D505	GENERAL	PROCESS DETAILS - PIPE PENETRATION
66	006M001	GENERAL	HVAC LEGENDS, NOTES, AND ABBREVIATIONS
67	006M501	GENERAL	HVAC DETAILS - DUCTS
68	006M502	GENERAL	HVAC DETAILS - DUCTS CONTINUED
69	006M503	GENERAL	HVAC DETAILS - GENERAL
70	007P001	GENERAL	PLUMBING LEGENDS, NOTES, AND ABBREVIATIONS
71	007P501	GENERAL	PLUMBING DETAILS - FACILITIES
72	007P502	GENERAL	PLUMBING DETAILS - FACILITIES CONTINUED
73	007P503	GENERAL	PLUMBING DETAILS - GENERAL
74	008E001	GENERAL	ELECTRICAL LEGEND - ONE-LINE, POWER, AND LIGHTING SYMBOLOGY
75	008E002	GENERAL	ELECTRICAL LEGEND - CONTROLS, COMMS, AUDIO, SECURITY AND SITE
76	008E501	GENERAL	ELECTRICAL DETAILS - UNDERGROUND CONDUIT
77	008E502	GENERAL	ELECTRICAL DETAILS - UNDERGROUND AND EQUIPMENT MOUNTING
78	008E503	GENERAL	ELECTRICAL DETAILS - LUMINAIRE MOUNTING
79	008E504	GENERAL	ELECTRICAL DETAILS - LUMINAIRE CONTROL SCHEMATICS
80	008E505	GENERAL	ELECTRICAL DETAILS - GROUNDING AND LIGHTNING PROTECTION
81	008E506	GENERAL	ELECTRICAL DETAILS - CABLE TRAY
82	008E507	GENERAL	ELECTRICAL DETAILS - SPECIAL SYSTEMS
83	008E508	GENERAL	ELECTRICAL DETAILS - SPECIAL SYSTEMS
84	008E509	GENERAL	ELECTRICAL DETAILS - TRANSFORMERS AND JUNCTION ENCLOSURE
85	008E510	GENERAL	ELECTRICAL DETAILS - GENERAL
86	009Y001	GENERAL	INSTRUMENTATION AND CONTROLS LEGEND
87	009Y002	GENERAL	INSTRUMENTATION AND CONTROLS LEGEND CONTINUED
88	009Y101	GENERAL	OVERALL FIBER PLAN
89	009Y501	GENERAL	INSTRUMENTATION DETAILS - FIBER
90	009Y502	GENERAL	INSTRUMENTATION DETAILS - SWITCHES AND TRANSMITTERS
91	009Y503	GENERAL	INSTRUMENTATION DETAILS - GENERAL
92	009Y620	GENERAL	TYPICAL CONTROL PANEL LAYOUT
93	009Y621	GENERAL	TYPICAL RIO PANEL LAYOUT
94	009Y622	GENERAL	TYPICAL CONTROL PANEL POWER SCHEMATIC
95	009Y630	GENERAL	TYPICAL DI WIRING SCHEMATIC
96	009Y631	GENERAL	TYPICAL DO WIRING SCHEMATIC
97	009Y632	GENERAL	TYPICAL AI WIRING SCHEMATIC
98	009Y633	GENERAL	TYPICAL AO WIRING SCHEMATIC
99	009Y634	GENERAL	TYPICAL INPUT/RELAY WIRING SCHEMATIC

REGISTERED PROFESSIONAL ENGINEER — CERTIFICATE NO. 75262 — GREGORY YOUNG — Date Signed 5-1-26 — ARIZONA, U.S.A.

REGISTERED PROFESSIONAL ENGINEER — CERTIFICATE NO. 41909 — RYAN RHOADES — Date Signed 5-1-26 — ARIZONA, U.S.A.

A/E FIRM	
PRIME:	HDR ENGINEERING, INC. DENVER, CO 80202
DESIGNED:	G. YOUNG
DRAWN:	N. LYNCH
TECH. REVIEW:	R. RHOADES
DATE:	AUG 2026
SUB SHEET NO.	000 G002
TITLE OF SHEET	GENERAL SHEET INDEX
	NORTH RIM - GRAND CANYON NATIONAL PARK
DRAWING NO.	113 182752
PMIS/PKG NO.	318719
SHEET	2 OF 99

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A/C	AIR CONDITIONING
A/E	ARCHITECT/ENGINEER
A	AMPERE
AB	ANCHOR BOLT
ABAN	ABANDON
ABC	AGGREGATE BASE COURSE
ABRO	AIRCRAFT BASED RADIO OPERATOR
ABT	ABOUT
AC	ALTERNATING CURRENT
ACK	ACKNOWLEDGE
ACP	ACOUSTIC CEILING PANEL, ASPHALTIC CONCRETE PAVEMENT
ACST	ACOUSTIC
AD	ADDENDUM, AREA DRAIN
ADDL	ADDITIONAL
ADH	ADHESIVE
ADJ	ADJUSTABLE, ADJACENT
AF	AMP FRAME, AMP FUSE
AFF	ABOVE FINISH FLOOR
AFG	ABOVE FINISH GRADE
AGGR	AGGREGATE
AI	AREA INLET, ANALOG INPUT
AIC	AMPS INTERRUPTING CAPACITY
ALIG	ALIGNMENT
ALT	ALTERNATE, ALTITUDE
ALUM	ALUMINUM
AM	ACUSTICAL MATERIAL
AMB	AMBIENT
ANC	ANCHOR
AO	ANALOG OUTPUT\
AP	ACCES PANEL
APRX	APPROXIMATE
APVD	APPROVED
ARCH	ARCHITECTURAL
ASSY	ASSEMBLY
AT	ACOUSTICAL TILE, AMP TRIP
ATC	ACOUSTICAL TILE CEILING
ATM	ATMOSPHERE AUTO AUTOMATIC
AUTO	AUTOMATIC
AUX	AUXILIARY
AVE	AVENUE
AVG	AVERAGE
AWG	AMERICAN WIRE GAGE
AWT	ACOUSTICAL WALL TILE
B TO B	BACK-TO-BACK
BAL	BALANCE
BBD	BULLETIN BOARD
BC	BASE CABINET, BOTTOM CHORD, BOLT CENTER, BOLT CIRCLE
BD	BOARD
BE	BOTH ENDS, BELL END
BF	BOTH FACES, BOTTOM FACE, BLIND FLANGE, BOARD FEET
BITUM	BITUMINOUS
BKG	BACKING
BL	BASE LINE
BLDG	BUILDING
BLK	BLOCK
BLKG	BLOCKING
BM	BENCHMARK, BEAM
BOC	BACK OF CURB
BOD	BOTTOM OF DUCT
BOG	BOTTOM OF GRILLE
BOL	BOTTOM OF LOUVER, BOLLARD
BOP	BOTTOM OF PIPE
BOR	BOTTOM OF REGISTER
BOT	BOTTOM
BOU	BOTTOM OF UNIT
BP	BASE PLATE
BRG	BEARING
BRGP	BEARING PLATE
BRKT	BRACKET
BS	BOTH SIDES
BTU	BRITISH THERMAL UNIT
BTW	BETWEEN
BTWLD	BUTT WELD
BU	BELL UP, BUILT-UP
BUR	BUILT-UP ROOFING
BW	BOTH WAYS
BYP	BYPASS
CTOC	CENTER TO CENTER
C&G	CURB AND GUTTER
C	CHANNEL SHAPE, CENTIGRADE, CONDUIT
CAB	CABINET
CAP	CAPACITY
CAT	CATALOG, CATALOGIORY
CAV	CAVITY
CB	CATCH BASIN
CCB	CONCRETE BLOCK
CCW	COUNTER CLOCKWISE
CDF	CONTROLLED-DENSITY FILL
CE	CONCRETE EDGE
CER	CERAMIC
CF	CUBIC FEET (FOOT)
CFL	COUNTER FLASHING
CHBD	CHALKBOARD
CHD	CHORD
CHFR	CHAMFER
CHH	COMMUNICATION HANDHOLE
CI	CURB INLET
CIP	CAST-IN-PLACE
CIPB	CONCRETE INTERLOCKING PAVER BALLAST
CIRC	CIRCULATION, CIRCULAR
CJ	CONSTRUCTION JOINT
CKT	CIRCUIT
CL	CENTERLINE, CLASS, CLOSE
CLG	CEILING
CLKG	CAULKING
CLR	CLEAR
CMH	COMMUNICATION MANHOLE
CMP	CORRUGATED METAL PIPE
CMU	CONCRETE MASONRY UNIT
CO	CLEANOUT, CONCRETE OPENING, CONTRACTING OFFICER
COL	COLUMN
COM	COMMON
COMB	COMBINATION
COMM	COMMUNICATION
COMP	COMPOSITION, COMPRESSIBLE, COMPOSITE
CON	CONCENTRIC
CONC	CONCRETE
CONN	CONNECTION
CONST	CONSTRUCTION
CONT	CONTINUOUS
COOR	COORDINATE
COR	CONTRACTING OFFICER REPRESENTATIVE
CORR	CORROSIVE, CORRUGATED
CP	CHECKER PLATE, CONTROL POINT
CPLG	COUPLING
CRL	CORROSION-RESISTANT LINING
CSC	COMPRESSION SLEEVE COUPLING
CSK	COUNTERSINK
CSS	CLINIC SERVICE SINK
CT	CERAMIC TILE
CTJ	CONTRACTION JOINT
CTR	CENTER
CTRL	CONTROL
CVT	CULVERT
CU	COPPER, CUBIC
CW	CLOCKWISE
CY	CUBIC YARD
d	PENNY (NAIL MEASURE)
D	DEEP, DIFFUSER, DRAIN
DB	DUCT BANK, DECIBEL, DRY BULB
DBA	DEFORMED BAR ANCHOR
DBL	DOUBLE
DC	DIRECT CURRENT
DEG	DEGREE
DEG C	DEGREE CENTIGRADE
DEG F	DEGREE FAHRENHEIT
DEMO	DEMOLITION
DEP	DEPRESSED
DEPT	DEPARTMENT
DET	DETAIL
DI	DROP INLET, DUCTILE IRON, DIGITAL INPUT
DIA	DIAMETER
DIAG	DIAGONAL, DIAGRAM
DIFF	DIFFERENTIAL, DIFFERENCE
DIM	DIMENSION
DISCH	DISCHARGE
DIST	DISTANCE, DISTRIBUTION
DIV	DIVISION
DL	DEAD LOAD
DMJ	DOUBLE MECHANICAL JOINT
DMPF	DAMP PROOFING
DN	DOWN
DO	DISSOLVED OXYGEN, DIGITAL OUTPUT, DITTO
DP	DEPTH
DPDT	DOUBLE POLE, DOUBLE THROW
DPST	DOUBLE POLE, SINGLE THROW
DS	DOWN SPOUT
DT	DOUBLE TEE, DRIP TRAP ASSEMBLY
DUP	DUPLICATE
DWG	DRAWING
DWL	DOWEL
DWR	DRAWER
E	EAST
EA	EACH, EXHAUST AIR
EC	ELECTRICAL CONTRACTOR
ECC	ECCENTRIC
ED	EQUIPMENT DRAIN
EDB	ELECTRICAL DUCT BANK
EE	EACH END
EF	EACH FACE
EFF	EFFLUENT, EFFICIENCY
EHH	ELECTRICAL HANDHOLE
EIFS	EXTERIOR INSULATION & FINISH SYSTEM
EJ	EXPANSION JOINT
EL	ELBOW, ELEVATION
ELEC	ELECTRICAL
EMBD	EMBEDDED
EMER	EMERGENCY
EMH	ELECTRICAL MANHOLE
ENCL	ENCLOSURE
ENGR	ENGINEER
ENTR	ENTRANCE
EOP	EDGE OF PAVEMENT
EQ	EQUAL
EQUIP	EQUIPMENT
EQUIV	EQUIVALENT
ES	EACH SIDE, EQUAL SPACE, EMERGENCY SHOWER
ESEW	EMERGENCY SHOWER AND EYE WASH
EST	ESTIMATE
EW	EACH WAY, EMERGENCY EYE/FACE WASH
EWC	ELECTRIC WATER COOLER
EWEF	EACH WAY, EACH FACE
EWTB	EACH WAY, TOP AND BOTTOM
EXC	EXCAVATION
EXH	EXHAUST
EXP	EXPANSION, EXPOSED
EXIST	EXISTING
EXT	EXTERIOR, EXTERNAL, EXTENSION
F TO F	FACE TO FACE
F&B	FACE AND BYPASS
FAB	FABRICATE
FB	FLOOR BEAM
FBD	FIBERBOARD
FBG	FIBERGLASS
FBM	BOARD FOOT MEASURE
FBO	FURNISHED BY OWNER
FC	FLUSHING CONNECTION
FCA	FLANGED COUPLING ADAPTER
FD	FLOOR DRAIN
FDC	FLEXIBLE DUCT CONNECTION
FDR	FEEDER
FDTN	FOUNDATION
FE	FLANGED END
FEC	FIRE EXTINGUISHER CABINET
FES	FLARED END SECTION
FEXT	FIRE EXTINGUISHER
FF	FAR FACE, FACTORY FINISH, FLAT FACE
FG	FINISHED GRADE
FH	FIRE HYDRANT
FIG	FIGURE
FIN	FINISH
FJT	FLUSH JOINT
FL	FLOW, FLOW LINE
FLEX	FLEXIBLE
FLG	FLANGE
FLOR	FLUORESCENT
FLR	FLOOR
FLS	FLASHING, FLUSH
FN	CORROSIVE, CORRUGATED
FO	FINISHED OPENING
FOB	FLAT ON BOTTOM
FOC	FACE OF CONCRETE, FACE OF CURB
FOF	FACE OF FINISH
FOM	FACE OF MASONRY
FOS	FACE OF STUDS
FOT	FLAT ON TOP
FO/E	FIBER OPTIC AND ELECTRIC
FPT	FEMALE PIPE THREAD
FR	FRAME
FRP	FIBERGLASS REINFORCED PLASTIC
FRTM	FIRE RETARDANT TREATED MATERIAL
FS	FLOOR SINK, FAR SIDE
FT	FEET, FOOT
FTG	FOOTING, FITTING
FUR	FURRED, FURRING
FURN	FURNITURE, FURNISH
FUT	FUTURE
FV	FACE VELOCITY
FW	FIELD WELD, FIRE WALL
FWD	FORWARD
FWE	FURNISHED WITH EQUIPMENT
FXTR	FIXTURE
G	GRILLE, GROUND
GA	GAUGE (METAL THICKNESS)
GAL	GALLON
GALV	GALVANIZED
GB	GRAB BAR, GRADE BREAK
GC	GROOVED COUPLING
GD	GUARD
GEN	GENERAL
GESC	GRADING, EROSION, AND SEDIMENT CONTROL
GFCI	GROUND FAULT CIRCUIT INTERRUPTER
GFMU	GROUND FACE MASONRY UNIT
GG	GUTTER GRADE
GJ	GROOVED JOINT
GL	GLASS
GLB	GLASS BLOCK, GLULAM BEAM
GND	GROUND
GP	GUY POLE
GR	GRADE
GRTG	GRATING
GSB	GYPSUM SHEATHING BOARD
GT	GREASE TRAP
GVL	GRAVEL
GW	GUY WIRE
GWB	GYPSUM WALLBOARD
GYP	GYPSUM HARDBOARD
H	HIGH
HB	HOSE BIBB
HBD	HARDBOARD
HC	HANDICAPPED, HOLLOW CORE, HORIZONTAL CURVE, HORIZONTAL CENTERLINE
HD	HEAD, HOT DIP
HDR	HEADER
HDW	HARDWARE
HEX	HEXAGONAL
HGL	HYDRAULIC GRADE LINE
HGR	HANGER
HH	HANDHOLE
HID	HIGH-INTENSITY DISCHARGE
HM	HOLLOW METAL
HORIZ	HORIZONTAL
HP	HIGH POINT, HORSEPOWER
HPC	HORIZONTAL POINT OF CURVATURE
HPIC	HIGH PERFORMANCE INDUSTRIAL COATING
HPS	HIGH-PRESSURE SODIUM
HPT	HORIZONTAL POINT OF TANGENCY
HR	HOSE REEL, HOUR
HS	HEADED STUD, HIGH STRENGTH
HSS	HOLLOW STRUCTURAL SHAPE
HT	HEIGHT
HTG	HEATING
HV	HIGH VOLTAGE
HVAC	HEATING, VENTILATING AND AIR CONDITIONING
HWD	HARDWOOD
HWL	HIGH WATER LEVEL
HYD	HYDRAULIC
HZ	HERTZ, CYCLES PER SECOND
ID	INSIDE DIAMETER, INTERIOR DIMENSION
IE	INVERT ELEVATION, FOR EXAMPLE
IF	INSIDE FACE
IH	INTAKE HOOD
IMP	IMPACT
IN	INCH
INC	INCLUDE, INCANDESCENT
INF	INFLUENT
INSTR	INSTRUMENTATION
INSUL	INSULATION
INT	INTERIOR, INTERSECTION
INTR	INTERMEDIATE, INTERIOR
INV	INVERT
IPS	IRON PIPE SIZE
IPT	INTERNAL PIPE THREAD
IR	INSIDE RADIUS, IRON ROD
IRR	IRRIGATION
ISO	ISOMETRIC
JB	JUNCTION BOX
JCT	JUNCTION
JF	JOINT FILLER
JST	JOIST
JT	JOINT
K	KIP
KB	KNEE BRACE
KCMIL	THOUSAND CIRCULAR MILS
KD	KNOCK DOWN
KO	KNOCK OUT
KSI	KIPS PER SQUARE INCH
KW	KILOWATT
L	ANGLE, LENGTH, LAVATORY, LINTEL
LAD	LADDER
LAM	LAMINATE
LATL	LATERAL
LB	LAG BOLT, POUND
LCTB	LIQUID CHALK AND TACK BOARD
LDG	LANDING
LDR	LEADER
LE	LIFTING EYE
LF	LINEAR FOOT
LG	LONG
LH	LEFT HAND
LIN	LINEAR
LIQ	LIQUID
LLH	LONG LEG HORIZONTAL
LLV	LONG LEG VERTICAL
LMLU	LIQUID MARKER LECTURE UNIT
LNG	LONGITUDINAL
LOC	LOCATION
LP	LOW POINT
LPS	LOW-PRESSURE SODIUM
LR	LONG RADIUS
LT	LEFT
LTD	LIMITED
LTG	LIGHTING
LTL	LINTEL
LTNG	LIGHTNING
LV	LOW VOLTAGE
LVL	LAMINATED VENEER LUMBER
LVR	LOUVER
LW	LIGHTWEIGHT
LWC	LIGHTWEIGHT CONCRETE
LWL	LOW WATER LEVEL
MA	MIXED AIR
MACH	MACHINED
MAINT	MAINTENANCE
MAN	MANUAL
MATL	MATERIAL
MAX	MAXIMUM
MB	MACHINE BOLT
MBR	MEMBER
MC	MECHANICAL CONTRACTOR, MECHANICAL COUPLING, MOMENT CONNECTION
MCB	METAL CORNER BEAD
MCJ	MASONRY CONTROL JOINT
MDMJ	MODIFIED DOUBLE MECHANICAL JOINT
MECH	MECHANICAL
MED	MEDIUM
MFR	MANUFACTURER
MH	MANHOLE, METAL HALIDE
MIN	MINIMUM
MIR	MIRROR
MISC	MISCELLANEOUS
MJ	MECHANICAL JOINT
ML	MASONRY LINTEL
MLO	MAIN LUGS ONLY
MMB	MEMBRANE
MO	MASONRY OPENING
MOD	MODULAR, MODIFY
MON	MONUMENT
MPT	MALE PIPE THREAD
MRGWB	MOISTURE-RESISTANT GYPSUM WALLBOARD
MS	MOP SINK
MSL	MEAN SEA LEVEL
MT	MOUNT
MU	MASONRY UNIT
MULL	MULLION
MV	MEDIUM VOLTAGE
MW	MONITORING WELL
N	NORTH, NEUTRAL
NA	NOT APPLICABLE
NAT	NATURAL, NATIONAL
NC	NORMALLY CLOSED
NEG	NEGATIVE
NF	NEAR FACE, NON-FUSED
NIC	NOT IN CONTRACT
NO	NORMALLY OPEN, NUMBER
NOM	NOMINAL
NPS	NOMINAL PIPE SIZE
NPT	NATIONAL PIPE THREAD
NS	NEAR SIDE
NPSHR	NET POSITIVE SUCTION HEAD REQUIRED
NTS	NOT TO SCALE
NWL	NORMAL WATER LEVEL
O TO O	OUT TO OUT
OA	OUTSIDE AIR, OVERALL
OC	ON CENTER
OCPD	OVER CURRENT PROTECTION DEVICE
OD	OUTSIDE DIAMETER
OED	OPEN END DUCT
OF	OUTSIDE FACE, OFFICE FURNISHING
OFCI	OWNER FURNISHED CONTRACTOR INSTALLED
OFOI	OWNER FURNISHED OWNER INSTALLED
OG	ORIGINAL GROUND
OH	OVERHEAD
OPNG	OPENING
OPGW	OPTICAL GROUND WIRE
OPP	OPPOSITE
OPT	OPTIONAL
OR	OUTSIDE RADIUS
ORD	OVERFLOW ROOF DRAIN
ORIG	ORIGINAL
OVFL	OVERFLOW
OVHG	OVERHANG
OZ	OUNCE
P	PAINT
PA	PUBLIC ADDRESS
PAR	PARALLEL, PARAPET
PB	PANIC BAR, PULL BOX
PBD	PARTICLE BOARD
PC	POINT OF CURVE, PIECE, PRECAST
PCC	POINT OF COMPOUND CURVATURE
PCF	POUNDS PER CUBIC FOOT
PCT	PERCENT
PDC	POWER DISTRIBUTION CENTER
PE	PLAIN END
PED	PEDESTAL
PEN	PENETRATION
PERF	PERFORATED
PERM	PERMANENT
PERP	PERPENDICULAR
PF	POWER FACTOR
PFMU	PREFACED MASONRY UNIT
PH	PHASE
PI	POINT OF INTERSECTION
PKG	PACKAGE
PL	PLATE, PROPERTY LINE, PRECAST LINTEL
PLAS	PLASTER
PLAT	PLATFORM
PLBG	PLUMBING
PLF	POUNDS PER LINEAR FOOT
PNEU	PNEUMATIC
POL	POLISH
POS	POSITIVE, POSITION
PP	POLYPROPYLENE, POWER POLE
PRC	POINT OF REVERSE CURVATURE
PREF	PREFINISHED
PREFAB	PREFABRICATED
PRELIM	PRELIMINARY
PREP	PREPARE
PRES	PRESSURE
PRI	PRIMARY
PROP	PROPERTY, PROPOSED
PROT	PROTECTION
PS	PIPE SUPPORT
PSF	POUNDS PER SQUARE FOOT
PSI	POUNDS PER SQUARE INCH
PSIA	POUNDS PER SQUARE INCH ABSOLUTE
PSIG	POUNDS PER SQUARE INCH GAGE
PST	PRESTRESSED
PT	POINT, POINT OF TANGENCY
PTN	PARTITION
PVC	POLYVINYL CHLORIDE, POINT OF VERTICAL CURVE
PVC-RGS	PVC COATED RGS
PVMT	PAVEMENT
PWD	PLYWOOD
PWJ	PLYWOOD WEB JOIST
PZ	PIEZOMETER
Q	RATE OF FLOW
QT	QUARRY TILE
QTR	QUARTER
QTY	QUANTITY
QUAL	QUALITY
R&R	REMOVE AND REPLACE
R&S	REMOVE AND SALVAGE
R	RADIUS, REGISTER, RISER
RA	RETURN AIR
RB	RESILIENT BASE, ROCK BERM
RCPT	RECEPTACLE
RD	ROOF DRAIN
REC	RECESS
RECD	RECEIVED
RECT	RECTANGULAR
RED	REDUCER
REF	REFERENCE
REINF	REINFORCING
REM	REMOVE
REQD	REQUIRED
RESIL	RESILIENT
RET	RETAINING, RETURN
REV	REVISION, REVERSE
RF	RESILIENT FLOORING
RFG	ROOFING
RFL	REFLECTED, REFLECTOR
RGH	ROUGH
RGS	RIGID GALVANIZED STEEL
RGS-PVC	PVC COATED RGS
RH	RELIEF HOOD, RIGHT HAND, RELATIVE HUMIDITY
RL	REQUIRED LAP
RLFA	RELIEF AIR
RND	ROUND
RNG	RUNNING
RO	ROUGH OPENING
ROW	RIGHT-OF-WAY
RPM	REVOLUTIONS PER MINUTE
RR	RAILROAD
RSP	ROCK SLOPE PROTECTION
RSOG	ROARING SPRINGS PUMPHOUSE
RT	RIGHT
RVT	RESILIENT VINYL TILE
RY	READY
S	SOUTH, SINK
SA	SUPPLY AIR
SAMU	SOUND-ABSORBING MASONRY UNIT
SAN	SANITARY
SB	SPLASH BLOCK
SC	SOLID CORE
SCH	SCHEDULE
SCHEM	SCHEMATIC
SCN	SCREEN
SE	STEEL/ALUMINUM EDGE
SEC	SECONDARY, SECONDS
SECT	SECTION
SEP	SEPARATE
SF	SQUARE FOOT, SILT FENCE
SG	SHEET GLASS, SEALANT GROOVE
SH	SHOWER
SHT	SHEET
SHTG	SHEATHING
SIL	SILENCE
SIM	SIMILAR
SJ	SLAB JOINT
SL	SLOPE, STEEL LINTEL
SLTD	SLOTTED
SLV	SLEEVE
SMLS	SEAMLESS
SOG	SLAB ON GRADE
SP	SOUNDPROOF, STANDPIPE
SPA	SPACING
SPEC	SPECIFICATION
SPLY	SUPPLY
SPST	SINGLE POLE SINGLE THROW
SPT	SET POINT
SQ	SQUARE
SR	SHORT RADIUS
SS	SERVICE SINK
SST	STAINLESS STEEL
ST	STREET
STA	STATION
STD	STANDARD
STIF	STIFFENER
STIR	STIRRUP
STL	STEEL
STOR	STORAGE
STR	STRUCTURAL, STRAIGHT
SUB	SUBSTITUTE
SUC	SUCTION
SUSP	SUSPENDED
SY	SQUARE YARD
SYM	SYMBOL
SYMM	SYMMETRICAL
SYN	SYNTHETIC
SYS	SYSTEM
T&B	TOP AND BOTTOM
T&G	TONGUE AND GROOVE
T	TILE, TREAD
TA	TOILET ACCESSORY, TEMPERED AIR
TAN	TANGENT
TBM	TEMPORARY BENCHMARK
TCE	TEMPORARY CONSTRUCTION EASEMENT
TCWL	TRANSCANYON WATERLINE
TEF	TROWELED EPOXY FLOORING
TEMP	TEMPORARY, TEMPERATURE
THD	THREAD
THK	THICK
THRESH	THRESHOLD
TKBD	TACK BOARD
TOB	TOP OF BOLT, TOP OF BANK, TOP OF BEAM, TOP OF BERM
TOC	TOP OF CURB, TOP OF CONCRETE
TOD	TOP OF DUCT
TOF	TOP OF FOOTING
TOG	TOP OF GRATING
TOL	TOLERANCE, TOP OF LEDGER
TOM	TOP OF MASONRY
TOP	TOP OF PLATE
TOPO	TOPOGRAPHY
TOS	TOP OF SLAB, TOP OF STEEL, TOE OF SLOPE
TOW	TOP OF WALL
TP	TOILET PARTITION, TELEPHONE POLE, TOE PLATE, TRAP PRIMER
TPD	TOILET PAPER DISPENSER
TPG	TOPPING, THROUGH PLATE GIRDER
TR	TRANSOM
TRANS	TRANSITION
TRD	TRENCH DRAIN
TYP	TYPICAL
U	URINAL
UG	UNDERGROUND
ULT	ULTIMATE
UNFN	UNFINISHED
UNO	UNLESS NOTED OTHERWISE
UTIL	UTILITY
V	VENT, VELOCITY, VOLT
VA	VOLT AMPERE
VAC	VACUUM
VAR	VARNISH, VARIABLE, VOLT AMPERES REACTIVE
VB	VAPOR BARRIER, VINYL BASE, VALVE BOX
VC	VERTICAL CURVE
VCP	VITRIFIED CLAY PIPE
VCT	VINYL COMPOSITION TILE, VERTICAL CENTERLINE
VEL	VELOCITY
VENT	VENTILATION
VERT	VERTICAL
VERTS	VERTICAL REINFORCING
VG	VERTICAL GRAIN
VIF	VERIFY IN FIELD
VIN	VINYL
VOL	VOLUME
VPC	VERTICAL POINT OF CURVATURE
VPI	VERTICAL POINT OF INTERSECTION
VPT	VERTICAL POINT OF TANGENCY
VS	VERSUS, VAPOR SEAL
VTR	VENT THROUGH ROOF
VWC	VINYL WALL COVERING
W/	WITH
W/O	WITHOUT
W	WATT, WEST, WIDE, WINDOW, WIRE, WIDE FLANGE BEAM
WB	WOOD BASE
WC	WATER CLOSET, WATER COLUMN
WD	WOOD, WIDTH
WF	WIDE FLANGE, WASH FOUNTAIN
WG	WIRE GLASS, WATER GAGE
WH	WALL HYDRANT, WEEP HOLE
WI	WROUGHT IRON
WL	WATER LEVEL
WLD	WELDED
WM	WIRE MESH
WP	WEATHERPROOF
WS	WATERSTOP, WATER SURFACE
WSCT	WAINSCOT
WSE	WATER SURFACE ELEVATION
WT	WEIGHT, WATER TIGHT
WTHP	WATERPROOF, WORKING POINT
WTP	WATER TREATMENT PLANT
WWF	WELDED WIRE FABRIC
WWTP	WASTEWATER TREATMENT PLANT
XP	EXPLOSION-PROOF
XS	EXTRA STRONG
XSECT	CROSS SECTION
XXS	DOUBLE EXTRA STRONG
YH	YARD HYDRANT
YS	YIELD STRENGTH

GENERAL NOTES

1. THESE ABBREVIATIONS APPLY TO THE ENTIRE SET OF CONTRACT DRAWINGS.
2. LISTING OF ABBREVIATIONS DOES NOT IMPLY THAT ALL ABBREVIATIONS ARE USED IN THE CONTRACT DRAWINGS.
3. ABBREVIATIONS SHOWN ON THIS SHEET INCLUDE VARIATIONS OF A WORD. FOR EXAMPLE, "MOD" MAY MEAN MODIFY OR MODIFICATION, "INC" MAY MEAN INCLUDED OR INCLUDING, AND "REINF" MAY MEAN REINFORCE OR REINFORCING.
4. SEE INSTRUMENTATION AND GENERAL LEGEND SHEETS FOR PROJECT-SPECIFIC EQUIPMENT AND PIPING SYSTEM ABBREVIATIONS.

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A/E FIRM
PRIME:
HDR ENGINEERING, INC.
DENVER, CO 80202

DESIGNED: G. YOUNG
CADD: N. LYNCH
TECH. REVIEW: M. COUGH/R. RHODES
DATE: MAY 2026

SUB SHEET NO. 000 G003

TITLE OF SHEET
GENERAL
ABBREVIATIONS

NORTH RIM - GRAND CANYON NATIONAL PARK

DRAWING NO. 113 182752
PMIS/PKG NO. 318719
SHEET 3 OF 99

MATERIALS IN PLAN/SECTION

FILLED REGIONS - MATERIALS
"MATERIAL" FILLED REGIONS ARE DRAFTING HATCH PATTERNS THAT REPRESENTS AN ACTUAL MATERIAL OR PRODUCT.

- MATERIAL - BITUMINOUS PAVING (ASPHALT)
- MATERIAL - CONCRETE - ARCHITECTURE PRECAST WALL PANEL
- MATERIAL - CONCRETE - ARCHITECTURAL PRECAST WALL PANEL
- MATERIAL - CONCRETE - CAST-IN-PLACE
- MATERIAL - CONCRETE - GLASS FIBER REINFORCED
- MATERIAL - CONCRETE - LIGHTWEIGHT
- MATERIAL - CONCRETE - STRUCTURAL
- MATERIAL - EFIS
- MATERIAL - FILL - COURSE - CRUSHED STONE
- MATERIAL - FILL - FINE - GRAVEL
- MATERIAL - FILL - FINE - SAND
- MATERIAL - FILL - MEDIUM - GRAVEL
- MATERIAL - FILTER POINT MAT - PLAN
- MATERIAL - FIREPROOFING - MINERAL WOOL
- MATERIAL - FIREPROOFING - SEALANT
- MATERIAL - FIREPROOFING - SPRAY-APPLIED
- MATERIAL - FLOOR - CARPET - STYLE 1
- MATERIAL - FLOOR - CARPET - STYLE 2
- MATERIAL - FLOOR - TERRAZZO - STYLE 1
- MATERIAL - FLOOR - TERRAZZO - STYLE 2
- MATERIAL - GLAZING
- MATERIAL - GLAZING - SPANDREL
- MATERIAL GRATING - PLAN
- MATERIAL GRATING - SECTION
- MATERIAL - GROUT
- MATERIAL - GYPSUM - PLASTER
- MATERIAL - GYPSUM - WALLBOARD
- MATERIAL - INSULATION - BATTING - SECTION
- MATERIAL - INSULATION - RIGID FOAM
- MATERIAL - MASONRY - BRICK
- MATERIAL - MASONRY - CMU
- MATERIAL - MASONRY - DOLMITE
- MATERIAL - MASONRY - MORTAR
- MATERIAL - MASONRY - STRUCTURAL CLAY TILE
- MATERIAL - METAL - ALUMINUM
- MATERIAL - METAL - CHECKERED PLATE - PLAN
- MATERIAL - METAL - ORNAMENTAL
- MATERIAL - METAL - STEEL
- MATERIAL - PLASTER STUCCO
- MATERIAL - PLASTIC
- MATERIAL - RESILIENT TILE
- MATERIAL - RIPRAP - PLAN AND/OR SECTION
- MATERIAL - SEALANT - ACOUSTICAL
- MATERIAL - SEALANT - FIRE SEALANT
- MATERIAL - SOD - SECTION
- MATERIAL - SOIL - COMPACTED - EARTH
- MATERIAL - SOIL - COMPACTED - FINE
- MATERIAL - SOIL - UNDISTURBED - FINE
- MATERIAL - SOIL - UNDISTURBED - EARTH
- MATERIAL - STONE - CAST
- MATERIAL - STONE - CUT FINISHED
- MATERIAL - SYSTEM - DRAINABLE EXTERIOR INSULATION AND FINISH
- MATERIAL - TERRAZZO - 1
- MATERIAL - TERRAZZO - 2
- MATERIAL - TILE - PORCELAIN
- MATERIAL - WEEP JOINT MORTAR PROTECTION SYSTEM - SECTION
- MATERIAL - WOOD - FINISH - COURSE GRAIN
- MATERIAL - WOOD - FINISH - FINE GRAIN
- MATERIAL - WOOD - FINISH VERTICAL GRAIN
- MATERIAL - WOOD FLOORING
- MATERIAL - WOOD - GLULAM
- MATERIAL - WOOD - MDF
- MATERIAL - WOOD - PARTICLE BOARD
- MATERIAL - WOOD - PLYWOOD
- MATERIAL - WOOD - CONTINUOUS - SECTION
- MATERIAL - WOOD - BLOCKING - SECTION

FILLED REGIONS - GRAPHICS
GRAPHIC FILLED REGIONS ARE DRAFTING HATCH PATTERNS THAT ARE A SIMPLE GRAPHIC PATTERN AND DOES NOT REPRESENT A MATERIAL OR PRODUCT.

- GRAPHIC - CONTRACT LIMIT - NOT-IN-CONTRACT
- GRAPHIC - DEMOLITION
- MATERIAL - RESILIENT SHEET

FILLED REGIONS - SURFACE
SURFACE FILLED REGIONS ARE MODELING HATCH PATTERNS THAT REPRESENT A MATERIAL OR PRODUCT AS REPRESENTED ON A VERTICAL OR HORIZONTAL SURFACE.

CEILING TILES

- SURFACE - CEILING TILE - 24" x 24"
- SURFACE - CEILING TILE - 24" x 24" - DEMO
- SURFACE - CEILING TILE - 24" x 48"
- SURFACE - CEILING TILE - 24" x 48" - DEMO
- SURFACE - CEILING TILE - 24" x 72"
- SURFACE - CEILING TILE - 24" x 72" - DEMO
- SURFACE - CEILING TILE - 48" x 48"
- SURFACE - CEILING TILE - 48" x 48" - DEMO
- SURFACE - CEILING - ACOUSTICAL TILE - SECTION

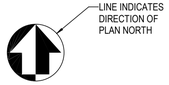
MASONRY

- SURFACE - MASONRY - BRICK - MODULAR - RUNNING BOND
- SURFACE - MASONRY - CMU - 8" x 16" - RUNNING BOND
- SURFACE - MASONRY - CMU - 8" x 16" - STACKED BOND

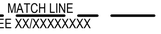
METALS

- SURFACE - ENTRANCE GRID - 65 MM

GENERAL SYMBOLOGY



NORTH ARROW



MATCH LINE



PLAN TITLE ON SHEET



DETAIL

FOR REFERENCING DETAILS INCLUDED IN DRAWING SET.
1. DETAIL NUMBER
2. SHEET WHERE DETAIL IS LOCATED *

SECTION

1. SECTION LETTER
2. SHEET WHERE SECTION VIEW IS FIRST CUT *

ELEVATION

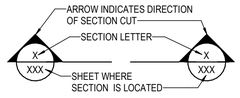
1. ELEVATION IDENTIFICATION NUMBER
2. SHEET WHERE POINT OF VIEW MARKER CAN BE FOUND *

SECTION, DETAIL, ELEVATION TITLE

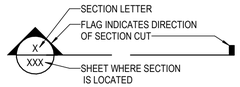


DETAIL CALLOUT

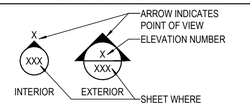
FOR REFERENCING DETAILS INCLUDED IN DRAWING SET



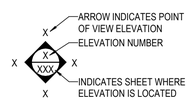
FULL BUILDING SECTION CUT MARKER



SECTION CUT MARKER



SINGLE ELEVATION OR PHOTO MARKER



MULTIPLE ELEVATION OR PHOTO MARKER



TARGET ELEVATION MARKER

* EXCEPTIONS WHERE SHEET NUMBER IS REPLACED BY A DASH (-)

1. FOR COMMON DETAILS, SECTIONS, ELEVATIONS OR DETAILS THAT ARE CUT OR CALLED OUT ON MULTIPLE SHEETS.
2. SECTIONS, ELEVATIONS OR DETAILS THAT ARE LOCATED ON THE SAME SHEET THEY ARE CUT OR CALLED OUT ON.

GENERAL TAGGING

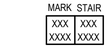
ROOM/SPACE NAME



ROOM OR SPACE



DOOR



FIRE DOOR/STAIR



NEW COLUMN GRID BUBBLE



EXISTING COLUMN GRID BUBBLE



WALL TYPE



CURTAIN WALL



WINDOW TYPE



ACCESSORY, FURNITURE, AND MISCELLANEOUS EQUIPMENT IDENTIFIER



LOUVER TYPE



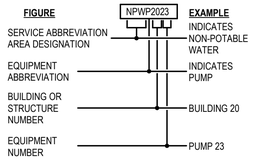
SHEET KEYNOTE



REVISION DELTA

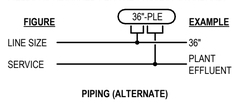
EQUIPMENT IDENTIFICATION

NOTE: THE BELOW EQUIPMENT TAG IS AN EXAMPLE ONLY. MODIFY FIELDS AS REQUIRED PER THE PROJECT STANDARD.

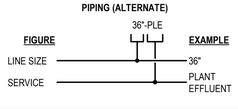


PIPING

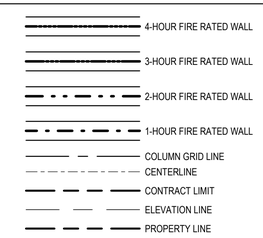
NOTE: THE BELOW PIPE TAG IS AN EXAMPLE ONLY. MODIFY FIELDS AS REQUIRED PER THE PROJECT STANDARD.



PIPING (ALTERNATE)



GENERAL LINE STYLES



SHEET NAMING CONVENTION

AREA DESIGNATION

GENERAL AREA DESIGNATIONS LISTED BELOW. LAST DIGIT OF AREA NUMBER DENOTES BUILDINGS AND SUB-AREAS

000	GENERAL SHEETS
010	PIPELINES - NORTH RIM
020	PIPELINES - NORTH RIM TO RSPH
030	PIPELINES - NORTH RIM TO SUPAI TUNNEL
040	PIPELINES - RSPH CAVE TO RSPH
050	ROARING SPRINGS PUMPHOUSE (RSPH)
060	NORTH RIM WATER TREATMENT
070	NOT USED
080	NORTH RIM WASTEWATER COLLECTION SYSTEM
090	POWER DISTRIBUTION - NORTH RIM
100	POWER DISTRIBUTION - NORTH RIM TO RSPH
110	POWER DISTRIBUTION - RSPH TO MANZANITA
120	POWER SYSTEM - COTTONWOOD CAMPGROUND

SINGLE DISCIPLINE DESIGNATOR & DISCIPLINE ORDER*

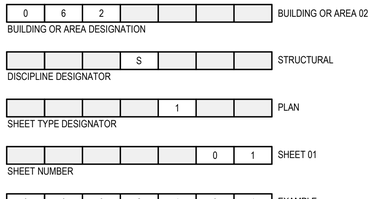
G	GENERAL
X	DEMOLITION/HAZARDOUS MATERIAL
V	SURVEYING/MAPPING
C	CIVIL
L	LANDSCAPE
S	STRUCTURAL
A	ARCHITECTURAL
F	FIRE PROTECTION
P	PLUMBING
D	PROCESS
M	MECHANICAL
E	ELECTRICAL
U	UNIVERSAL/OTHER DISCIPLINES
Y	INSTRUMENTATION & CONTROLS

SHEET TYPE DESIGNATOR

0	GENERAL (SYMBOLS, LEGENDS)
1	PLANS
2	ELEVATIONS
3	SECTIONS
4	LARGE SCALE VIEWS
5	DETAILS
6	SCHEDULES AND DIAGRAMS
8	PROFILES
9	3D REPRESENTATIONS

EXAMPLE 1 (WITH BUILDING SERIES, SINGLE LETTER DESIGNATOR)

NORTH RIM WATER TREATMENT STRUCTURAL FOUNDATION PLAN, DRAWING 01



GENERAL NOTES

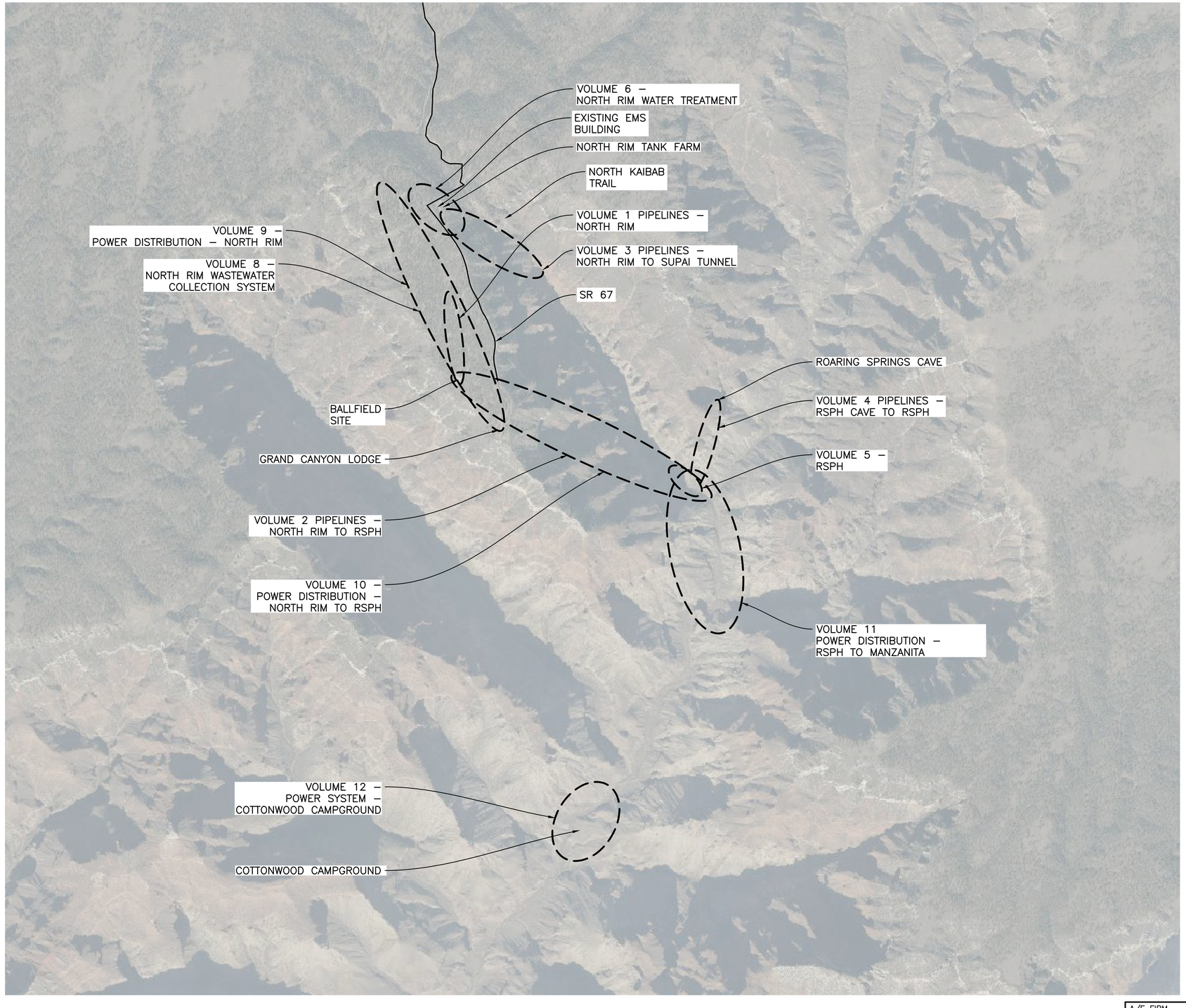
A. THIS IS STANDARD SHEET SHOWING COMMON SYMBOLOGY. ALL SYMBOLS ARE NOT NECESSARILY USED ON THIS PROJECT.

B. SCREENING OR SHADING OF WORK IS USED TO INDICATE EXISTING COMPONENTS OR TO DE-EMPHASIZE PROPOSED IMPROVEMENTS TO HIGHLIGHT SELECTED TRADE WORK. REFER TO CONTEXT OF EACH SHEET FOR USAGE.

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REGISTERED PROFESSIONAL ENGINEER
CERTIFICATE NO.
41859
RYAN RHODES
Date Signed 5-1-26
ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG CADD N. LYNCH TECH. REVIEW: M. COUCH/R. RHODES DATE: MAY 2026	SUB SHEET NO. 000 G004	TITLE OF SHEET GENERAL GENERAL LEGEND NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 4 OF 99



WORK PHASES SUMMARY

REFER TO SPECIFICATION 01 11 00 – SUMMARY OF WORK FOR ADDITION DETAILS. THE WORK SHALL BE CONDUCTED WITH THE FOLLOWING PHASING REQUIREMENTS:

1. VOLUME 01 PIPELINES – NORTH RIM
   1.1. THE BALLFIELD SITE PREPARATION AND BALLFIELD ACCESS ROAD SHALL BE COMPLETE PRIOR TO VOLUME 02 BORE MOBILIZATION.

2. VOLUME 04 PIPELINES – ROARING SPRINGS CAVE TO ROARING SPRINGS PUMPHOUSE
   2.1. THE EXISTING ROARING SPRINGS CAVE PIPELINE MUST REMAIN OPERATIONAL WITH A MAXIMUM CONTINUOUS SHUTDOWN DURATION OF 10 DAYS UNTIL COMPLETION OF THE TRANS–CANYON WATERLINE PROJECT.

3. VOLUME 05 – ROARING SPRINGS PUMPHOUSE
   3.1. THE ROARING SPRINGS PUMPHOUSE PUMP/ELECTRICAL WORK CAN NOT OCCUR UNTIL THE VOLUME 1 AND 2 PIPELINES WORK HAS BEEN COMPLETED, TESTED, AND COMMISSIONED AS THE NEW PUMPS ARE NOT FUNCTIONAL WITH THE OLD PIPING. THE ROARING SPRINGS PUMPHOUSE CAN ONLY BE SHUT DOWN AFTER CLOSURE OF THE NORTH RIM TO VISITORS NOVEMBER 15 AND PRIOR TO OPENING MAY 15.

4. VOLUME 10 – POWER DISTRIBUTION – NORTH RIM TO ROARING SPRINGS PUMPHOUSE
   4.1. OVERHEAD POWER REPLACEMENT CAN ONLY OCCUR WHEN THE ROARING SPRINGS PUMPHOUSE IS OFFLINE FOR THE WINTER SEASON. THIS GENERALLY OCCURS NOVEMBER 15 THROUGH MAY 15 (DOES NOT HAVE TO BE DONE THE SAME CONSTRUCTION SEASON AS THE REHABILITATION OF THE ROARING SPRINGS PUMP STATION, CAN OCCUR BEFORE, DURING OR AFTER VOLUME 5 WORK).

1500 0 10 3000
SCALE OF FEET
1" = 1500'

4/23/2026 Young, Gregory C:\Users\gryoung\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\GEN\000-G005.dwg

ARIZONA BOARD OF TECHNICAL REGISTRATION CERTIFICATE NO. 41809 RYAN RHOADES Date Signed 2-27-26 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG CADD C. WITTENMEIER TECH. REVIEW: M. GOUCH/R. RHODES DATE: MAY 2026	SUB SHEET NO. 000 G005	TITLE OF SHEET GENERAL PROJECT AREAS AND VICINITY MAP NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 5 OF 99